



Graduation Project — Task Documentation

Person 3 | Anomaly Detection + Alerts

Backend • Frontend • AI

Powered by LLM + RAG + Agentic AI

Overview

You are the watchdog of InsightX AI. After a report is confirmed by the Manager, your system wakes up, compares the new numbers against historical data and thresholds, and fires alerts to the Owner when something is wrong. You also run a daily scheduled job.

Backend Tasks

ASP.NET Core | Background Services | Semantic Kernel

1. Database Tables to Create

Table	Key Columns
Alerts	Id, CompanyId, DepartmentId, KPISName, CurrentValue, Threshold, Message, Rec...

2. Anomaly Detection Logic (3 Levels)

Task	Details
Level 1 — Threshold	If current value crosses the company KPI threshold → fire alert immediately. Example: defect_rate = 7% but threshold = 5% → ALERT.
Level 2 — Z-Score	Compare current month to same month last year. If difference > 20% → fire alert. Example: March this year 600 units vs March last year 1000 units → ALERT.
Level 3 — Trend	If same KPI drops for 3 consecutive months → fire alert. Example: Jan 1000, Feb 900, Mar 800 → downward trend ALERT.

3. API Endpoints

Task	Details
GET /alerts	List all alerts for current company. Filter by seen/unseen.

Task	Details
PUT /alerts/{id}/seen	Mark alert as seen by Owner.
POST /alerts/run	Manually trigger anomaly detection for a specific report.

4. Scheduled Daily Job

Implement a Background Service in ASP.NET Core that runs every day at 8 AM. It loops over all companies, fetches their latest metrics from ExtractedMetrics table, and runs the 3-level detection logic. Any anomaly found creates a new row in Alerts table.

AI Tasks — Alert Message Generation

Semantic Kernel | Prompt Engineering

After anomaly is detected, call the LLM to generate a human-readable alert message and a recommendation. Use this prompt:

```
You are a business analyst. KPI: {{kpi_name}} Current value:
{{current_value}} Expected (threshold): {{threshold}} Historical context:
{{history}} Write ONE short alert message (max 20 words). Then write ONE
short recommendation (max 25 words). Return JSON: {"message": "...",
"recommendation": "..."}

```

Example Output

```
{"message": "Production dropped 40% in March vs threshold of 1000 units.",
"recommendation": "Investigate raw material supply chain and schedule urgent
supplier meeting."}

```

Frontend Tasks

Task	Details
Alerts Page	List all alerts with color coding: Red = unseen, Gray = seen. Show message + recommendation
Notification Bell	In the navbar: show count of unseen alerts. Click to open alerts drawer.
Alert Card	Each alert shows: KPI name, current value, threshold, message, recommendation, date.

Example Alert in Database

Field	Value
KPIName	daily_production
CurrentValue	600
Threshold	1000
Message	Production dropped 40% in March.
Recommendation	Contact suppliers immediately.
SeenByOwner	false

